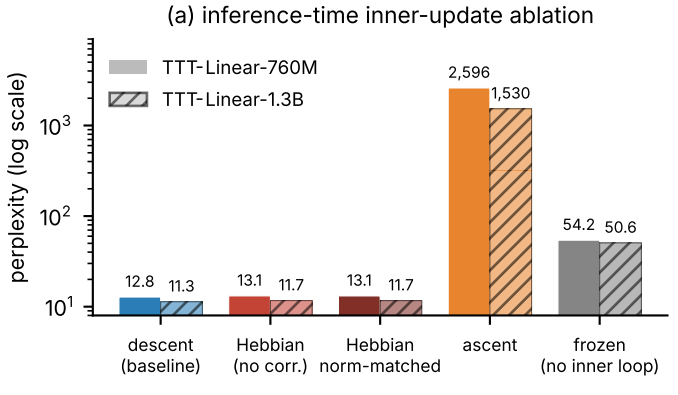
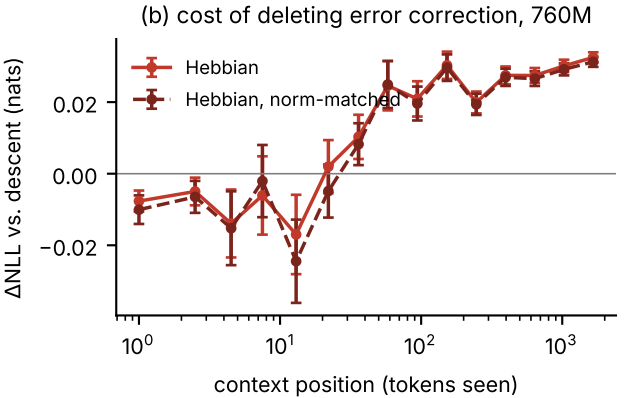

(a) inference-time inner-update ablation
TTT-Linear-760M
TTT-Linear-1.3B
perplexity (log scale)
10^3
10^2
10^1
12.8
11.3
13.1
11.7
13.1
11.7
2,596
1,530
54.2
50.6
descent (baseline)
Hebbian (no corr.)
Hebbian norm-matched
ascent
frozen (no inner loop)


(b) cost of deleting error correction, 760M
Hebbian
Hebbian, norm-matched
ΔNLL vs. descent (nats)
0.02
0.00
−0.02
10^0
10^1
10^2
10^3
context position (tokens seen)